

Generation Interconnection System Impact Study Report

For Queue Project Q15-0434

Trinidad 138 kV
September 29, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0434. The Interconnection Customer proposes a 83.6 MW battery energy storage facility interconnecting at the Trinidad 138 kV substation in the Virginia area. The request was received on February 13, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 214 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,937,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CHANCELLOR ROW / BEAR ISLAND 345 KV
Contingency: LEAVITT / BIRCHWOOD 345 KV
Loading: 111.2% (Rating: 1,376 MVA)
DFAX: 19.5%
Pre-project loading: 110.0%
2. Facility: BIRCHWOOD / NAUVOO 345 KV Line
Contingency: FREDERICKSBURG / DUNN SIDING 345 KV
Loading: 110.0% (Rating: 1,663 MVA)
DFAX: 15.4%
Pre-project loading: 109.2%
3. Facility: BELTSVILLE / BUNKER HILL 138 KV
Contingency: GLENANNAN / TABER 345 KV L
Loading: 109.2% (Rating: 812 MVA)
DFAX: 5.3%
Pre-project loading: 108.6%
4. Facility: BUNKER HILL / NASA 138 KV line
Contingency: GLENANNAN / TABER 345 KV
Loading: 108.1% (Rating: 855 MVA)
DFAX: 5.3%
Pre-project loading: 107.5%
5. Facility: BEAR ISLAND / DUNN SIDING 345 KV
Contingency: BIRCHWOOD / LEAVITT 345 KV
Loading: 105.4% (Rating: 1,885 MVA)
DFAX: 19.7%
Pre-project loading: 104.6%
6. Facility: MECKLENBURG POWER STA / BEAR 345 KV
Contingency: CLOVER / BEAR 345 KV
Loading: 104.6% (Rating: 1,927 MVA)

DFAX: 32.5%

Pre-project loading: 103.2%

7. Facility: LEAVITT / BIRCHWOOD 345 KV L

Contingency: BER ISLAND / DUNN SIDING 345 KV Line

Loading: 104.1% (Rating: 2,046 MVA)

DFAX: 21.8%

Pre-project loading: 103.2%

8. Facility: FREDERICKSBURG / DUNN SIDING 345 KV line

Contingency: BIRCHWOOD / NAUVOO 345 KV

Loading: 103.2% (Rating: 1,345 MVA)

DFAX: 12.5%

Pre-project loading: 102.4%

9. Facility: MCCARTNEY / NORTH TOWER TAP 345 KV L

Contingency: BEAR / MECKLENBURG POWER STATION 345 KV

Loading: 102.9% (Rating: 1,435 MVA)

DFAX: 11.2%

Pre-project loading: 102.3%

10. Facility: N TOWER TAP / HAGGIN 345 KV

Contingency: MECKLENBURG POWER STATION / BEAR 345 KV

Loading: 101.1% (Rating: 1,281 MVA)

DFAX: 11.4%

Pre-project loading: 100.4%

11. Facility: BEAR / FLUVANNA P.S 345 KV

Contingency: LEAVITT / BIRCHWOOD 345 KV

Loading: 101.1% (Rating: 1,169 MVA)

DFAX: 21.5%

Pre-project loading: 99.5%

12. Facility: RIVA RD / SHINGLETOWN 345 KV

Contingency: TABER / GLENANNAN 345 KV L

Loading: 100.2% (Rating: 1,356 MVA)

DFAX: 7.1%

Pre-project loading: 99.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHANCELLOR ROW / BEAR ISLAND 345 KV	Rebuild line with higher-capacity conductor	\$20,561,000	2% (\$469,000)
BIRCHWOOD / NAUVOO 345 KV Line	Rebuild line with higher-capacity conductor	\$153,427,000	2% (\$3,101,000)
BELTSVILLE / BUNKER HILL 138 KV	Reconductor line	\$237,000	3% (\$7,000)
BUNKER HILL / NASA 138 KV line	Reconductor line	\$631,000	3% (\$19,000)
BEAR ISLAND / DUNN SIDING 345 KV	Reconductor line	\$19,329,000	2% (\$380,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MECKLENBURG POWER STA / BEAR 345 KV	Reconductor line	\$55,641,000	4% (\$2,314,000)
LEAVITT / BIRCHWOOD 345 KV	Reconductor line	\$49,722,000	2% (\$1,055,000)
FREDERICKSBURG / DUNN SIDING 345 KV line	Reconductor line	\$20,041,000	2% (\$405,000)
N TOWER TAP / MCCARTNEY 345 KV Line	Reconductor line	\$49,076,000	5% (\$2,290,000)
NORTH TOWER TAP / HAGGIN SUBSTATION 345 KV	Reconductor line	\$17,421,000	5% (\$901,000)
FLUVANNA P.S / BEAR 345 KV line	Reconductor line	\$61,973,000	2% (\$1,524,000)
SHINGLETOWN / RIVA RD 345 KV	Reconductor line	\$24,238,000	2% (\$472,000)
	Total allocated to project		\$12,937,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.